

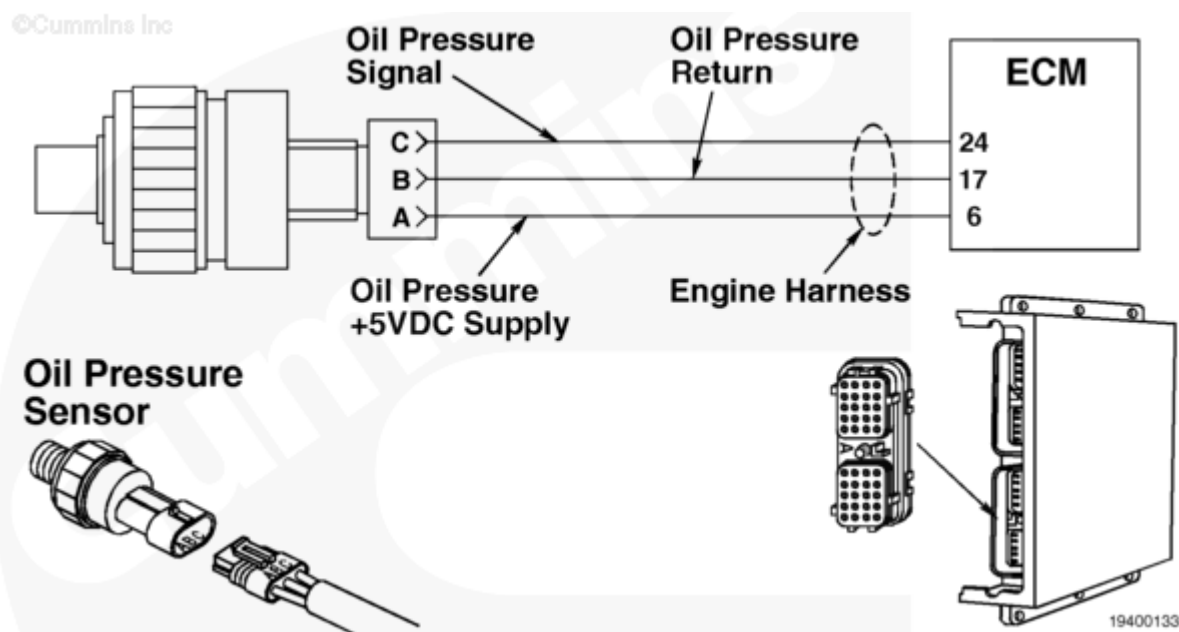
Fault Code: 415

Oil Pressure - Engine Protection

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Overview

Codes	Reason	Effect
Fault Code: 415 PID(P): P100 SPN: 100 FMI: 1 Lamp: Engine Protection SRT: 00-367	Very low oil pressure has been detected. Voltage signal at oil pressure signal pin 24 of the engine harness indicates oil pressure lower than 83 kPa [12 psi] at 600 rpm, 110 kPa [16 psi] at 800 rpm, 138 kPa [20 psi] at 1500 rpm, and 172 kPa [25 psi] above 2100 rpm.	Calibration-dependent progressive power and speed derate and engine shutdown with increasing time after the alert.



Oil Pressure Sensor Circuit

Circuit Description

The oil pressure sensor is used by the ECM to monitor the lubricating oil pressure. The ECM monitors the voltage on the signal pin and converts this to a pressure value. The oil pressure value is used by the ECM for the engine protection system.

Component Location

The oil pressure sensor is located on the engine block, toward the front of the ECM.

Shoptalk

Possible causes of this fault code include:

- Malfunctioning or damaged engine oil pressure sensor
- Low oil level
- External oil leaks
- Plugged lubricating oil filters
- Contaminated oil
- Oil aeration
- High oil temperature
- Malfunctioning or damaged main oil rifle pressure regulator
- Malfunctioning or damaged piston cooling nozzle
- Malfunctioning or damaged oil suction tube
- Malfunctioning or damaged oil transfer plumbing
- Malfunctioning or damaged oil pump
- Malfunctioning or damaged lubricating pump high pressure relief valve
- Internal engine damage
- Malfunctioning or damaged oil cooler element

Refer to Troubleshooting Fault Code t05-415 (</qs3/pubsys2/xml/en/procedures/19/19-t05-415.html>)